



Mitigating distortions in cast automotive subframes: A finite element simulation approach

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Abstract

Ensuring dimensional accuracy is critical for automotive parts, as they must meet strict tolerances - often within a few hundred microns at key locations to ensure proper assembly within the vehicle. However, the casting and heat treatment of large aluminium structures introduces thermal stresses and distortions that must be controlled to maintain these requirements. To address this challenge, a mathematical modeling based framework has been developed to predict and mitigate the deformations by designing the casting process along with the heat-treatment and quenching process. The simulations conducted, using an in-house developed Finite Element based solver, are validated against physical test data from industrial production. The comparisons show that the models developed in this work, along with their underlying assumptions, provide a comprehensive representation of the thermo-mechanical behaviour of subframe parts during the casting process. Finally, further refinements to the modelling approach have been suggested to improve the test-simulation correlations.

Keywords Aluminium automotive subframes · Modelling · Distortions · Finite element method · Validation · Casting process

1 Introduction

Fuel efficiency plays a pivotal role in achieving sustainability goals by directly reducing energy consumption, greenhouse gas emissions, and resource depletion. Industries face a significant challenge due to increasing demands on fuel efficiency and several innovations are centred around this goal. The automotive industry needs to balance between fuel efficiency, vehicle performance and safety crashworthiness during the design of new products. Replacement of steel and cast iron by light metal alloys has gained momentum leading to a large number of parts being manufactured with lighter metals [1, 2]. However changing to lighter metals requires the parts to withstand stringent requirements of strength and durability along with the stresses induced during casting of large structural parts. Aluminium alloys are extensively used in the production of large-scale, complex

cross-sectional architectural and industrial profiles due to its moderate strength, excellent plasticity, good castability and weldability, and strong corrosion resistance. The complex cross sections from the casting of aluminium often leads to parts with varying thickness, shape and inability to cool evenly inside cross sections and ultimately causing non-uniform cooling during the quenching process [3]. This non-uniform cooling causes large thermal gradients, high residual stresses and thermally induced distortions, such as warping and twisting [4]. The outcome is products not meeting the dimensional requirements for automotive parts, defects due to creep in the cast parts or cracking of the parts during quenching [5]. It is imperative to understand the behaviour of cast aluminium parts undergoing heat treatment after casting. The residual stresses and distortion in cast aluminium components during quenching are influenced by the quenching rate and the degree of temperature non-uniformity within the casting, which is directly linked to the heat transfer characteristics of aluminium castings during the quenching process. Unlike gas and air quenching where the heat transfer is relatively straightforward and the heat transfer coefficient remains constant, quenching in fluids (oil or water) involves three stages of heat transfer.

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The first stage, referred to as film boiling or vapour blanket formation, occurs at high temperatures [6, 7], where fluid comes into direct contact with the hot part surface. During this phase, rapid localized boiling leads to the development of a vapour film (e.g., steam) around the surface of the hot component, and heat is transferred to the fluid through this vapour layer. The second stage, referred to as nucleate boiling is characterized by a collapse of the steam layer. In this stage, a nucleate boiling regime forms, involving complex phenomena such as bubble nucleation, growth, and detachment. During the bubble growth, significant heat is transferred from the hot surface to the surrounding liquid and the bubble growth being the main heat transfer mechanism [7]. The final stage, known as convective cooling; occurs when the surface temperature falls below the fluid's boiling point. Heat transfer at this stage is primarily through direct convection to the fluid [2]. The phase transformation and nucleation growth involved in these steps makes modelling the quenching process a difficult task for researchers.

Clean and cheap water quenching instead of high pollution and high cost oil quenching or polymer aqueous solution quenching has become a trend in the heat treatment industry with increased application in the casting of automotive parts. However, the design of heat treatment processes including the process parameters has largely been reliant on experiments rather than modelling and simulation practices. Experimental measurements face three key limitations: (1) for large and complex components, measuring internal stress as a function of depth is both challenging and time-intensive; (2) in most cases, cracking in quenched components is caused by transient stress during quenching, but experiments can only measure the final internal (residual) stress, not the transient stress; and (3) understanding the origins of complex quenching stresses is difficult without the support of computer simulations for stress analysis [8].

The article is arranged as follows: the next section describes the modelling methodology adopted to represent the solidification and heat treatment procedure in the casting facility followed by the results and discussion from the simulations along with the limitations conducted and possible next steps to improve the research.

2 Modelling methodology

As discussed in Section 1, the modelling and simulation of solidification and heat treatment of cast parts faces challenges from representing the entire process within the simulation to predict the distortions in the cast part. In this work, a FEM based thermo-mechanical simulation model has been developed to define the process and validated against data

from tests. The entire CAD setup was meshed to capture the geometry of the parts and the supports; an in-house solver StaMiSim (Stress and Microstructure Simulator) developed internally by the authors was used to simulate the complex process ([9–11]).

The mechanical analysis is carried out in the fully solid parts of the cast product as well as in the coherent part of the mushy zone. The upper boundary of the coherent mushy zone corresponds to a solid volume fraction equal to 0.9 in this study with the A356 alloy. For a complete description of the mechanical model employed, see [12] and [13]. In mushy-zone mechanics, the choice of a solid fraction threshold for coherency, typically $g_s = 0.9$, is based on experimental and modeling evidence that at this level the dendritic network becomes sufficiently interconnected to transmit mechanical loads. Below this threshold, the solid skeleton lacks rigidity and behaves more like a suspension, while above it, the structure can sustain stresses similar to a porous solid. Here only a brief description is provided while a complete list of nomenclature is given in Table 3. Denoting the solid stress tensor σ_s , the mixture momentum balance in the coherent part of the mushy zone can be expressed by:

$$\nabla \cdot [g_s \sigma_s] - \nabla [g_l p_l] + [\rho_s g_s + \rho_l g_l] \mathbf{g} = 0 \quad (1)$$

where $\mathbf{g}$ is the gravity vector. The model formulation has been based on the classical small deformation theory where the total strain ϵ , derivable from the displacement field $\mathbf{u}$ is subdivided into viscoplastic, elastic and thermal components.

$$\epsilon = \frac{1}{2}(\nabla \mathbf{u} + [\nabla \mathbf{u}]^T) = \epsilon^{vp} + \epsilon^e + \epsilon^T \quad (2)$$

The thermal strain component can be written as an integral of the (volumetric) thermal expansion coefficient β_T .

$$\epsilon^T = -\frac{1}{3} \int_T^{T_{coh}} \beta_T(T) dT \cdot \mathbf{I} \quad (3)$$

The constitutive equation for elasticity (Hook's law) defines how the stress depends on the elastic strain ϵ^e and the temperature dependent Young's modulus and Poisson's ratio entering the matrix $\mathbf{D}$.

$$\sigma = \mathbf{D}(T) \cdot \epsilon^e \quad (4)$$

Equation 1 simplifies to the usual Cauchy's equation for a fully solid material

$$\nabla \cdot \sigma_s + \rho_s \mathbf{g} = 0 \quad (5)$$

A set of equations assuming steady state creep above a temperature T_0 and work hardening below this temperature, are applied for viscoplastic deformation in the fully solidified regime:

$$\bar{\sigma} = F(T) \cdot (\phi_0 + \phi)^{n(T)} \cdot (\dot{\varepsilon}_p)^{m(T)} \quad (6)$$

$$d\phi = \begin{cases} d\bar{\varepsilon}_p & \text{when } T \leq T_0 \\ 0 & \text{when } T > T_0 \end{cases} \quad (7)$$

where $F(T)$, $n(T)$, $m(T)$ are tabulated functions of temperature. ϕ accounts for work hardening and is the integrated value of the viscoplastic strain at temperatures below T_0 . For the A356 alloy, the elastoviscoplastic material parameters, including the hardening behavior described by Eqs. 6 and 7, were calibrated using the experimental flow stress curves reported in [14]. The fitted values for the A356 material is shown in Table 1 and based on the curve fitting the parameter T_0 was set to 420°C. This value is taken from fitting of model parameters to experiments on a similar AlSiMg alloy. In this case a test sample was subject to continuous straining and cooling from above 500°C to 80°C in a Gleeble® machine at different strain rates. As a very good agreement was achieved, it demonstrated the ability of our simple model to capture the main effects of temperature and strain histories on the build-up of stresses during cooling. This type of experiment and fitting has been reported in [15].

The mechanical model for Eqs. 1–5 describes stress and strain evolution in the fully solid regions and the coherent mushy zone during alloy solidification. The mushy zone is treated as a two-phase mixture of solid and liquid, and the governing equations are based on classical continuum mechanics under small deformation theory. The following assumptions apply: the material is isotropic, strains are small so linearized kinematics are valid, the liquid phase is incompressible, and turbulence-induced pressure fluctuations are neglected. Starting from the mixture momentum

balance, Eq. 1 expresses equilibrium of forces in the mushy zone, where the divergence of solid stress is weighted by solid fraction, the gradient of liquid pressure is weighted by liquid fraction, and body forces from gravity acting on both phases are summed to zero under quasi-static conditions. The strain tensor in Eq. 2 is derived from the displacement field and decomposed into elastic, thermal, and viscoplastic components. Thermal strain Eq. 3 accounts for volumetric expansion integrated over temperature, while Eq. 4 applies Hooke's law for isotropic elasticity, with temperature-dependent elastic properties. When the material is fully solid ($g_s = 1$), Eq. 1 simplifies to the standard Cauchy equilibrium Eq. 5. These relations provide the foundation for modeling residual stresses and deformation during solidification and subsequent cooling.

To further enhance predictive capability, the model could incorporate the Bauschinger effect, which accounts for yield asymmetry during load reversal caused by internal back stresses from dislocation structures and precipitate interactions, by introducing a kinematic hardening component into the constitutive law. This approach aligns with the need for advanced constitutive formulations capable of capturing non-linear plasticity and stress redistribution under cyclic loading, as emphasized by Perzyk in his assessment of constitutive equations for cooling castings and deformation modeling over wide temperature ranges [16, 17]. This addition captures the stress redistribution and plastic strain evolution under cyclic or reversed loading conditions which are critical for processes such as straightening, thermal cycling, and distortion control in Al–Si–Mg alloys. However, this effect has been excluded in the present computational model based on certain simplifying assumptions, including monotonic loading and isotropic hardening behavior. After solution heat treatment at approximately 540 °C, most alloying elements besides Si in A356 are dissolved into the aluminum matrix, resulting in a supersaturated solid solution upon quenching. At this stage, there are no significant Mg_2Si precipitates to act as strong obstacles to dislocation motion. Based on similar observations by [18], consequently, the Bauschinger effect is present but less pronounced because internal back stresses primarily arise from dislocation substructures and interactions with eutectic Si phase rather than precipitate-dislocation interactions, which dominate in the aged (T6) condition. For this reason as well as lack of test data quantifying this effect, the current model assumes negligible kinematic hardening and does not account for yield asymmetry during load reversal, focusing instead on monotonic deformation during solidification and cooling. The implications of bi-directional loading and the consecutive influence of the Bauschinger effect on stress-strain behavior are analyzed further in Section 3.

Table 1 Empirical model settings for the A356 flow stress

Temp (°C)	$F(T)$ (MPa)	$n(T)$	$m(T)$
0	22.0	0.3	0.0
50	21.5	0.3	0.0
100	21.0	0.3	0.0
150	20.0	0.3	0.006
200	18.9	0.27	0.016
250	17.5	0.21	0.04
300	16.4	0.15	0.1
350	16.0	0.125	0.15
400	15.6	0.02	0.19
450	12.3	0.0	0.22
500	7.2	0.0	0.25
550	2.0	0.0	0.277

2.1 Modelling as-cast deformation

As-cast (AC) deformation is in this context the geometric deviations from the CAD geometry for the automotive part immediately after the casting process and before the heat treatment process. The casting process includes the solidification in a mould, cooling after ejection from the mould, sand core knock-out and removal of the feed gates. Generally, we will then have a deviation from the detailed CAD geometry. Remark that the deviations do not only appear from thermal gradients and internal stresses developed during solidification, cooling and degating, but also from the mould distortions. The mould will be hot and thereby thermally expanded when the liquid metal solidifies against the mould walls. The mould distortions are typically the main cause of the AC deformation.

The thermal analysis in casting involves a detailed description of the melt flow and all the cooling channels, inserts and coatings in the mould tool by CastCad (Simulation Driven Casting Design & Optimization Software) [19]. The mould design has to allow for the different regions where reduced cooling or heating is necessary to avoid porosity, and for enhanced cooling in other regions, and at the same time allow for as small deflections as possible. As the inner side is hotter than the outer side the general behaviour of the upper and lower mould will be to curve upward and downward at the edges so that a larger gap evolves there which can lead to flashing (melt leakage). This will also lead to a thicker automotive part locally at this place, compared to the cold mould cavity.

The mechanical analysis includes both the cast part and the equipment around the cast, with the lowermost part

being the machine plate over the crucible and the uppermost the pressurized tilt plate. Figure 1 shows an example. The equipment includes a baseplate within the machine plate with holes for the fill-tubes to deliver liquid metal into the feeding gate system and finally into the mould cavity. There is a lower and an upper mould enclosing the cavity, but through these two mould parts there are also inserts that may be heated or cooled dependent on the filling strategy. Between the base plate and the lower mould there are supports to reduce the lower mould deflection and similarly there are rods between the upper mould and the plates above to reduce the upper mould deflection. To stabilize the two mould parts there are also guides and keys with tolerances allowing for thermal expansion while at the same time keeping the mould parts in the same horizontal position. The computational mesh includes therefore in total about 30 different domains (dependent on the actual casting part) with mechanical boundaries between each domain which in some cases allows for air-gaps, in other cases gaps within set tolerances or with no gaps (bolted). Capturing the boundary conditions is critical to represent the casting process in the simulation. On the free boundaries there are no constraints on the displacements while for the other boundaries spring forces in tangential and normal directions are applied with a constant stiffness or with a variable stiffness depending on the displacement. With a variable stiffness one is able to have a very high stiffness against another material if penetration starts to occur while at the same time allow air-gaps to form by reducing the stiffness when small gaps form. The boundary conditions are then the result of an iterative procedure where the stiffness may change during the iteration process. A table showing the normal variable stiffness that

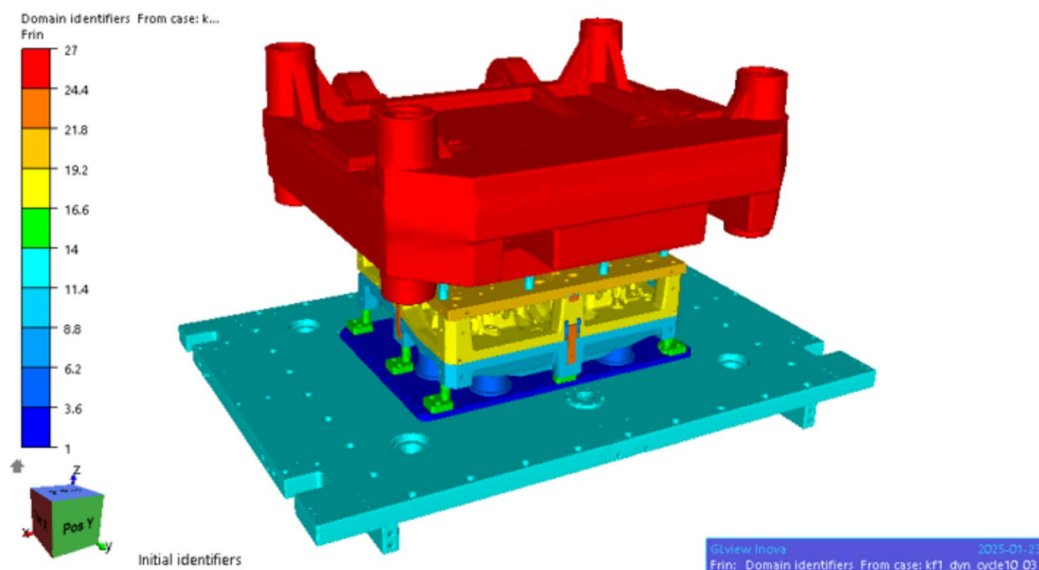


Fig. 1 Example with in total 27 different finite element domains which are used to model as cast deformation. 3.0 Million elements were used in the analysis

Table 2 Displacement in meter vs. Stiffness in Pa/m

Displacement (m)	Stiffness (Pa/m)
-0.10	1×10^5
-0.00020	5×10^5
-0.00008	5×10^6
-0.00005	5×10^7
-0.00002	5×10^8
-0.00001	1×10^9
0.00000	1×10^{10}
0.00001	1×10^{10}
0.00002	1×10^{11}
0.00003	1×10^{12}
0.00005	1×10^{13}

allows airgap development between two steel parts while at the same time does not allow significant penetration is shown in Table 2. The tangential stiffness has been set to 1×10^8 Pa/m. A schematic representation of the casting cell is shown in Fig. 2, and a drawing of the machine without the mould system (but including the machine plate) is shown in Fig. 3.

The equipment domains are modelled as thermoelastic materials (different types of steels), while the cast part including the gates is modelled as a thermo-elastovisco-plastic material with flow stress governed by Eq. 6. In each mechanical time step, the strains and stresses in the fully solid material and the coherent mushy zone are first evaluated. Next, the displacement field and strains in the liquid

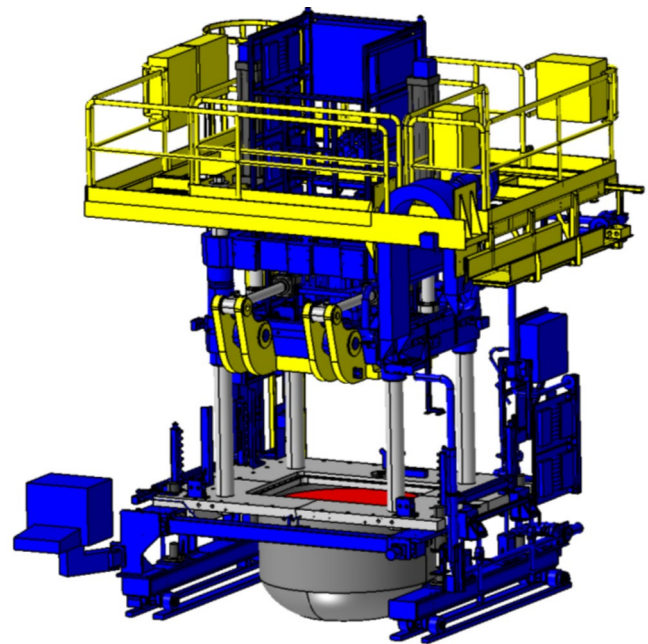


Fig. 3 Drawing of the external equipment outside the computational domain, but including the machine plate above the crucible (melt in crucible is marked with red colour) and the tilt plate and frame above

part is computed by treating this domain as elastic, and by using the displacements in the solidified part and in the adjacent equipment part as Dirichlet boundary conditions. When the material in an element (i.e. an element within the finite element numerical representation) is predicted to solidify, the stress in the element is given a zero value by assigning it an initial plastic strain corresponding to

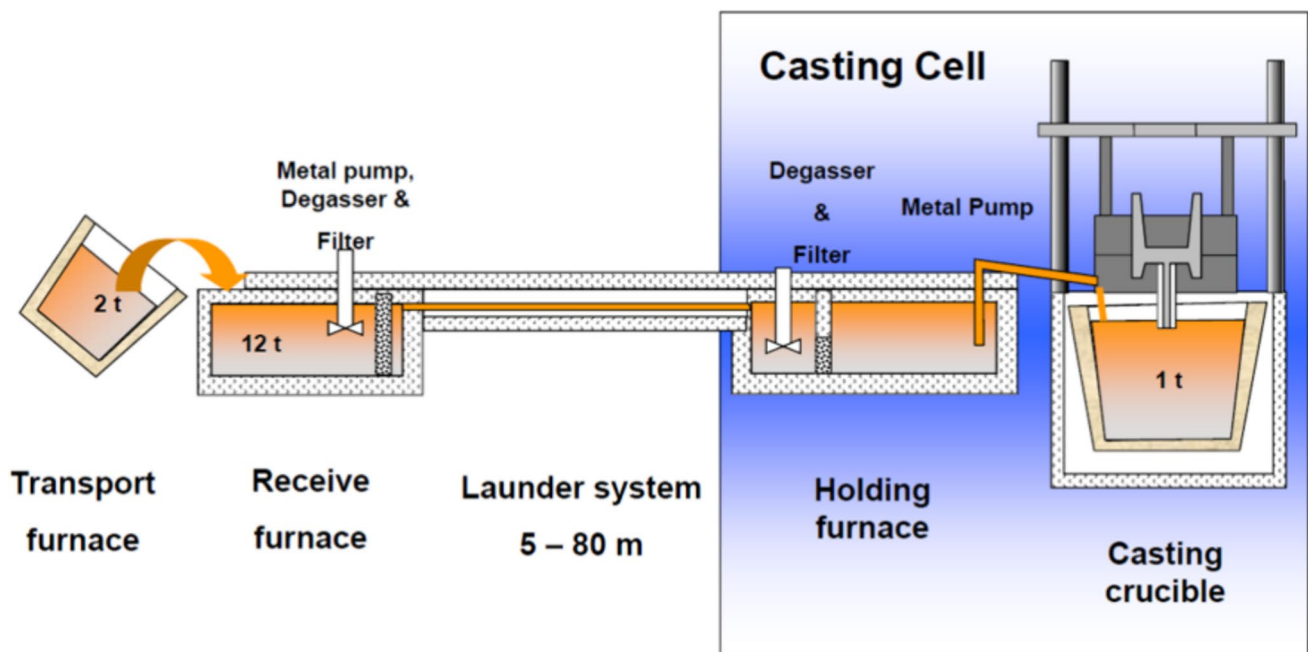


Fig. 2 Schematic representation of the casting cell at Aludyne Farsund

its total strain derived from the displacement field in the liquid domain. This element will then fit into the already deformed part of the cast domain and it will also fit into the deformed walls of the mould or the inserts that makes the boundaries for the cast material. The cold mould cavity in the simulation is expanded typically 0.5 to 1 % from the centre of gravity of the cast part compared to the CAD drawing. So the cold mould expansion in addition to the hot mould thermal expansion at the moment a surface element enters the coherent semi-solid material will make the initial shape of the cast part. The cast part will then be exposed to the mould movements (as the mould cools down) and in addition the cast part will shrink. After the opening of the mould the cast part will cool down further by air cooling before de-gating and final cooling down to room temperature. The simulated AC deformation can then be quantified as any deviation from the CAD-drawing. The final AC deformation is then the results of several processes; the cold mould cavity expansion, the heated mould (amount of heating will depend on the cast cycle number), mould deformation, air-cooling with and without a fan, sand core removal and finally de-gating by sawing off the gates which will result in a sudden stress relief and a change of the final deformation state.

2.2 Modelling the deviations from the nominal geometry based on a choice of contact points

The AC deformation is not the same as the deformation of the final product. Further distortions or warpings will be introduced during T6 heat treatment applied to achieve optimum strength of casting. Deformation will take place both during solution heat treatment by creep mechanisms and especially during the subsequent water quenching. Also during the artificial ageing process applied for the A356 alloy some distortions may appear. The distortions of the final product are therefore the sum of the AC distortions and the heat treatment (HT) distortions, as shown by Eq. 8.

$$d_{N,product} = d_{N,AC} + d_{N,HT} \quad (8)$$

At any time during the process chain the deviations from the CAD can be measured, in order to understand where the most important distortions will be made. For example in the coordinate measuring machine (CMM) a probe will move against the surface, if the probe hits the metal surface too early (compared to the CAD surface) the deviation is positive. If the probe hits the metal surface after the CAD surface the deviation is negative. When the deformed part is positioned in the CMM it will slightly translate and rotate the part compared to the initial state in the model formulation. This is because the contact points (locator points) between cast part and CMM will also be deformed, but inside the machine these contact points will be assigned zero deviation from the CAD. We therefore have to compensate for the rotation and then calculate a new normal displacement from the nominal geometry while at the same time adjust for the cold mould expansion. We can then directly compare with the measurements from the CMM. Figure 4 shows the deviation from the nominal geometry from a particular example after heat treatment.

2.3 Modelling of distortions during heat treatment including quenching

After casting, a part is put in a rack together with several other parts and heated up to 540°C for several hours. Depending on the positioning in the rack and the weight of the part some creep may appear due to gravity acting on the part which is supported by 3 or 4 points based on the design of the rack. Figure 5 shows an example of a cast part inside the rack.

As the rack is moved out of the furnace it is air-cooled for some few seconds, and it is then submerged into a tank for water quenching. The heat transfer in the water tank is calculated from correlations for convection, nucleate boiling, critical heat flux, transition boiling and film boiling.

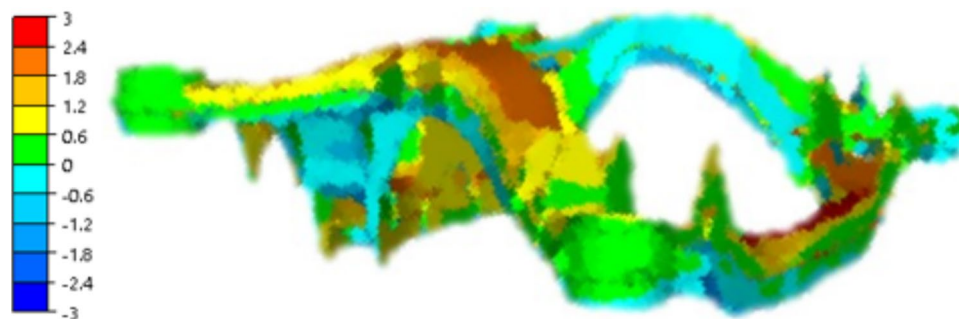


Fig. 4 Simulated displacements from the nominal plotted on the surface of an automotive aluminium subframe. Values are in millimeters. Positive values are outward movement compared to the CAD, while negative values are inward movement compared to the CAD drawing.

The diagonal distance for this part is about 1.3 meter. The colour scale is from -3mm to +3mm in steps 0.6mm from dark blue via green and yellow to orange and red

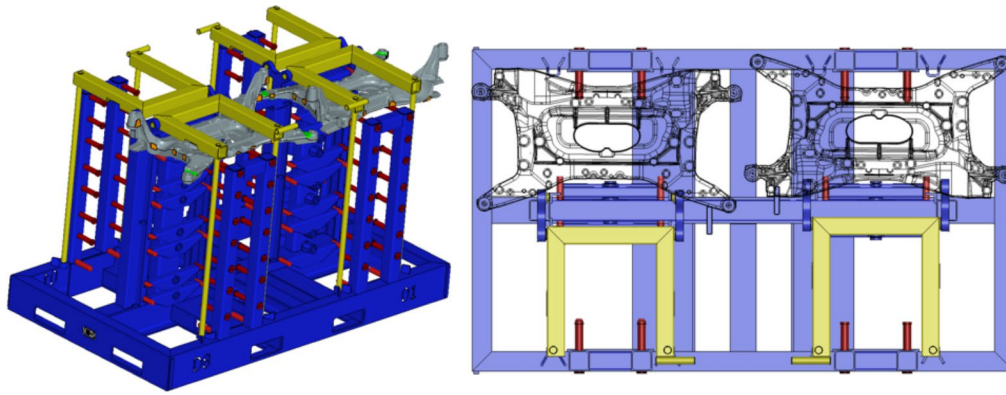


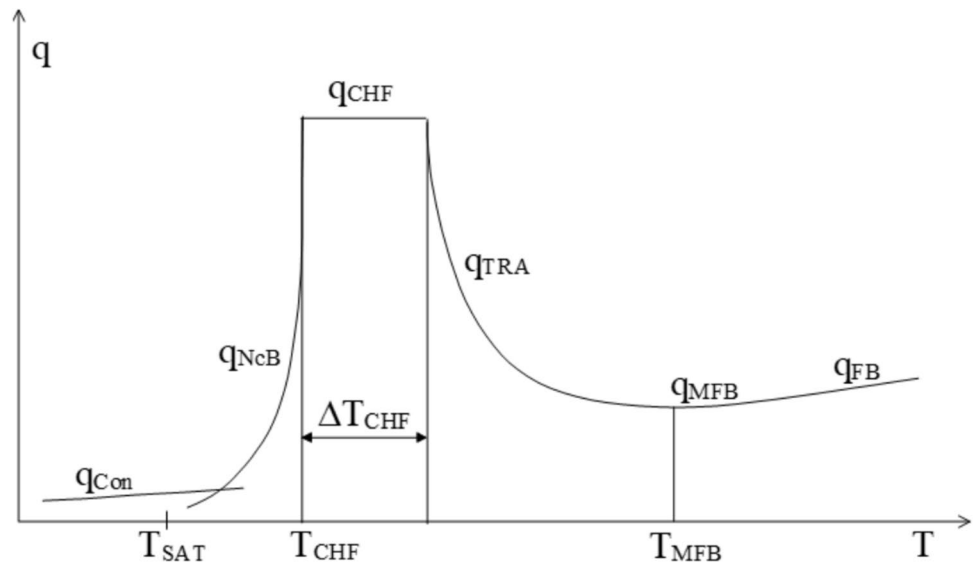
Fig. 5 Example with 32 parts stacked in a steel rack

Figure 6 shows these different heat transfer correlations that are applied by StaMiSim. For temperatures below and to some extent over the boiling temperature T_{SAT} , the water cools the surface by convection calculated by a correlation for q_{CON} . When the surface temperature of the cast part is well over the boiling temperature, nucleate boiling q_{NcB} is the dominating heat transfer mechanism. The heat transfer increases very rapidly with the temperature until the vapour close to the surface restricts the bubble nucleation, and we are in the critical heat flux area q_{CHF} . The amount of vapour at the surface increases and the transition heat flux q_{TRA} decreases as the surface is nearly covered by vapour which acts as an insulator. At the end of the transition area we reach a minimum of the heat flux at the minimum film boiling temperature T_{MFB} , which is also called the Leidenfrost temperature. Above T_{MFB} the film boiling heat flux q_{FB} increases with the surface temperature due to enhanced radiation and conduction in the vapour film. The vapour film is not stable and the correlations of the critical heat flux, transition boiling and film boiling are dependent on the water sub-cooling and the relative velocity between

water and cast part. The water sub-cooling is the temperature difference between the boiling temperature T_{SAT} and the water temperature.

When the rack moves into the tank the lowermost layers of cast parts in the rack will move for a longer time in the water than the uppermost parts which will only move for a short time before the parts are fully submerged into the water. This is taken into account in the model by defining the position of the cast part inside the rack. The velocity of the rack which acts as forced convection cooling in the water is used in the water-cooling correlations. It has been observed that the rack velocity may have a considerable influence on the deformations induced by the quenching. As the part temperature initially is close to 540 °C the cooling will start in the film boiling regime, with reduced heat transfer. It will take time before the surface temperature is reduced enough to enter the more efficient nucleate boiling regime. Due to the different thickness of the automotive part some areas will cool down much faster than others, and this creates large thermal gradients and thereby large thermally induced stresses. It is therefore imperative to model the

Fig. 6 Water cooling curve with correlations that are used in the StaMiSim modelling tool. It shows the heat density as function of surface temperature on the cast part



water-cooling as accurately as possible, including the transition from film boiling to nucleate boiling, in order to estimate the quench distortions. The water cooling correlations are as defined in [20] but with modified coefficients reflecting the water tank. Along with this, the vapour or steam bubble generation by boiling and the temperature increase in the water poses significant variations in the conditions between the different layers in the tank. For the temperature increase in the tank an experiment with thermocouples at several locations in the tank has been used to make an empirical connection between rack and parts weights and a position and time dependent temperature increase in the water. The resulting water temperature is used to calculate the water subcooling locally and time dependent all over the surface of the cast part. The subcooling enters the water-cooling correlations and will therefore have an impact on the resulting quench deformations. Figure 7 shows the measured water temperature increase for some points in the rack (same rack as shown in Fig. 5).

As the measurement points are at the sides and edges of the rack an ad hoc assumption is that inside the rack and in between the parts the water temperatures will start to increase immediately with no time delay.

The boiling of the water creates for a short time a large volume of vapour bubbles in the water tank. The vapour content will be higher for the uppermost layers than for the lowermost layers as the vapour bubbles will rise from the lower layers of the tank through the higher layers against the surface. The boiling in the tank is significant, cascades of water reaches the level of 10–20 cm above the still water tank level. In principle it would be possible to model the whole rack with computational fluid dynamics including the vapour bubbles, but considering the number of parts in the

rack, the complicated shape of the parts including the cavities, and the travelling of all the steam against the surface, this is not doable in practice. Therefore, a fully empirical approach is followed here.

To take the vapour content into consideration a factor to reduce the calculated heat transfer coefficient is introduced. This factor varies between 0 (no vapour bubbles) and 1 (fully covered by vapour bubbles). At the bottom of the first layer, it is assumed no vapour as the vapour will be released upwards. As we move upwards in the tank more and more vapour will be accumulated the higher we are. However, after some few seconds the boiling will be reduced so the vapour factor is both dependent on time and the vertical position in the tank (similar to the water temperature). A fully empirical approach was employed to determine the vapour content correction factor by analysing deformation after quenching across various automotive part types. The layer variations of each part were used to calibrate the correction factor, ultimately selecting the best fit with the measured deformations. The factor was then scaled based on the mass of the rack and parts, using the reference value shown in Fig. 8. Notably, the time scale represents the local time for a specific point in the rack, starting from the moment it is submerged beneath the water tank's surface.

The orientation of the part in the water also needs to be taken into account, as a flat wall facing downwards or upwards will experience a different natural convection flow pattern compared to a sloped or vertical wall. Similarly, cavities inside the automotive part needs special treatment as the cooling will be reduced inside the cavity due to entrapment of air and vapour. To identify whether an element surface in the FE mesh is inside a cavity or not, a ray beam technique was used. 5 rays are sent out from an element

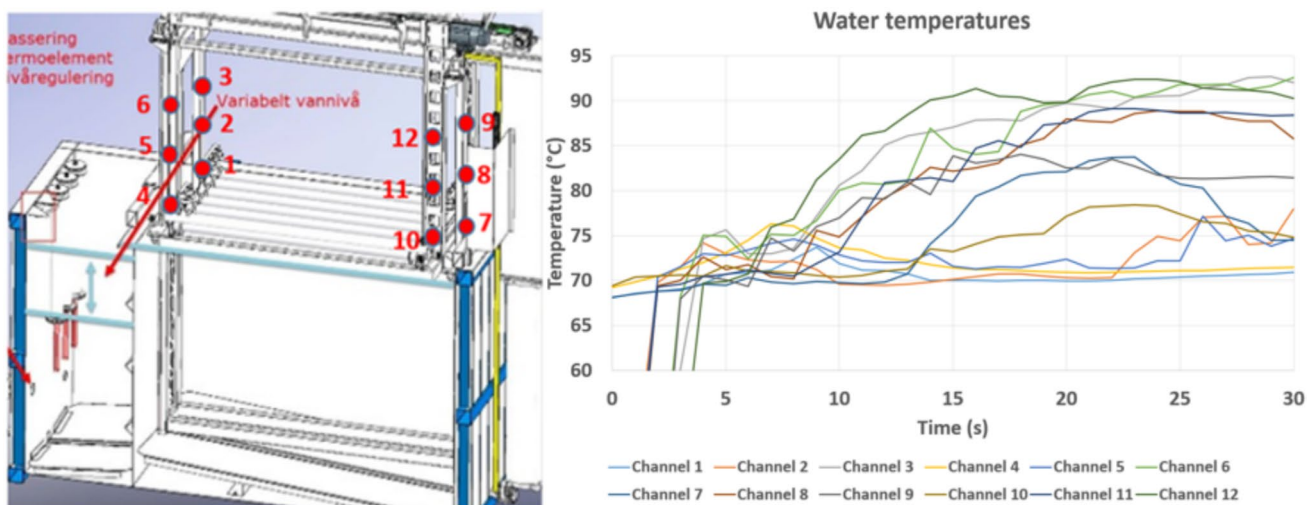
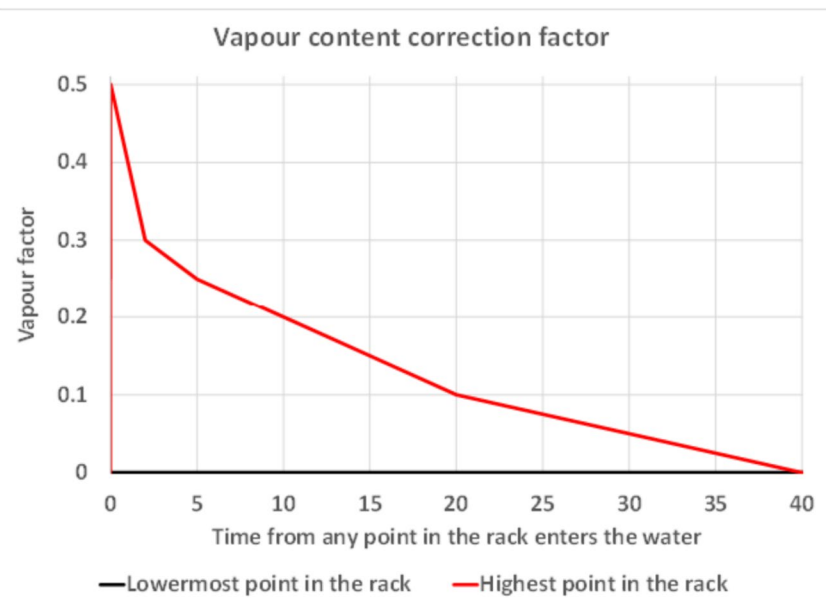


Fig. 7 Measured water temperatures in the water tank used to generate a general correlation for water tank temperatures dependent on time and depth in the tank. Time zero is the time when the lowermost part

in the rack enters the water tank. Time scale is from zero to 30 seconds and temperature scale from 60 to 95°C

Fig. 8 Vapour content correction factor which is scaled with rack mass. Time is running from the moment a location in the rack is submerged into the water. For the lowermost point in the rack the vapour correction factor is zero



surface, vertically and at an angle with a tangent on the surface. If all the rays find an element during the search along the defined length of the rays, a cavity is assumed. Figure 9 shows the marker for a cavity calculated on a cross-member.

Inside the cavity it is reasonable to assume that the “roof” section of the cavity experiences reduced cooling as air or vapour can be caught at least for a small amount of time (cooling down to the water temperature is typically done in 1-5 seconds). Also on the outer side of the part horizontally oriented surfaces will experience reduced cooling. It has been found, by comparing with experimental results from several automotive parts, that the angle dependent cooling

also depends on the vertical position in the rack. A plausible explanation for this is that the flow gets very chaotic, and includes large amount of vapour due to the boiling, the higher we are in the rack. Therefore, high in the rack, it is more a general reduction in the overall cooling imposed by the vapour content correction factor shown in Fig. 8. Figure 10 shows the external angle dependent factors from the bottom to the top of the rack. The angle-dependent factors are interpolated between the top and bottom. It’s important to note that these factors apply to a part moving through water. For a part that has a flat bottom (with its outer surface normal aligned with gravity), effective cooling still occurs because

Fig. 9 Marker on cut-plane through a cross-member beam with internal cavity. Element surfaces coloured red have been detected to be inside the cavity

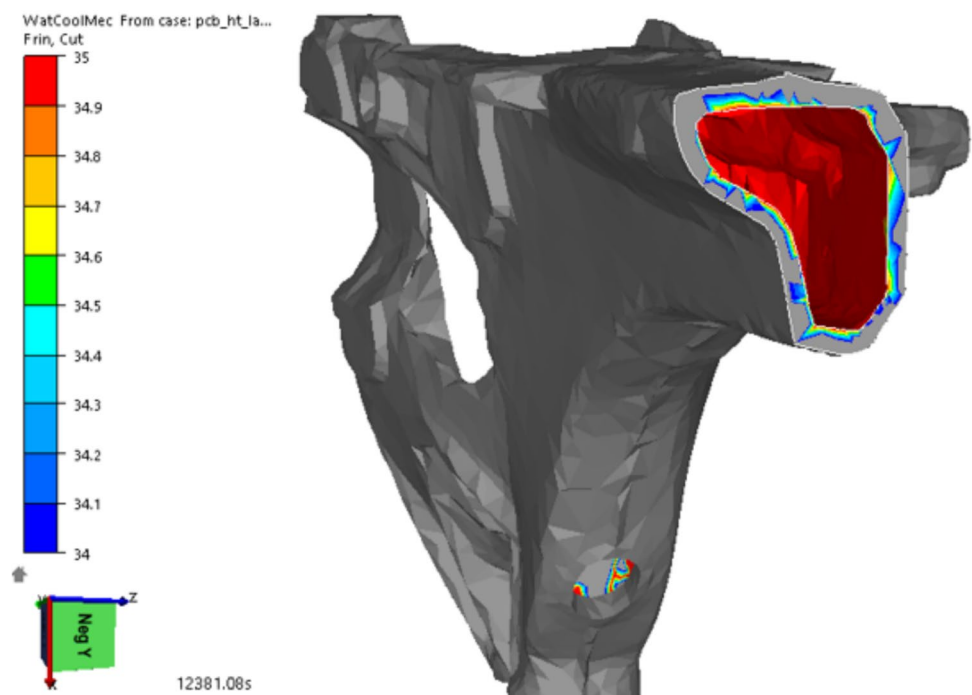
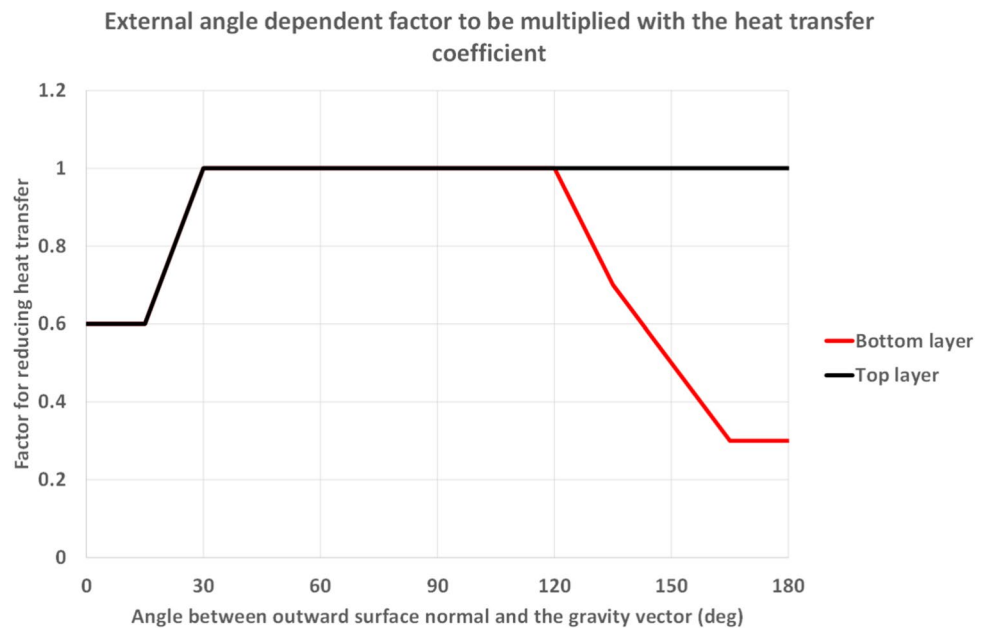


Fig. 10 External (i.e. not inside cavities) angle dependent factor to be multiplied with the heat transfer coefficient from the vapour modified water cooling correlations



the movement forces the water away. However, the cooling is less effective than sloped or vertical walls where the water more easily moves away from the downward moving part. Similarly, Fig. 11 shows the internal cavity angle dependent factors from the bottom through the middle and onto the top of the rack. For layers in between the bottom and the top of the rack the angle dependent factors are interpolated.

The models described above are simulated with StaMi-Sim to capture the distortions in the casting process. The subsequent section reports the numerical simulation results and their validation against experimental data acquired from industrial casthouse trials for the following cases:

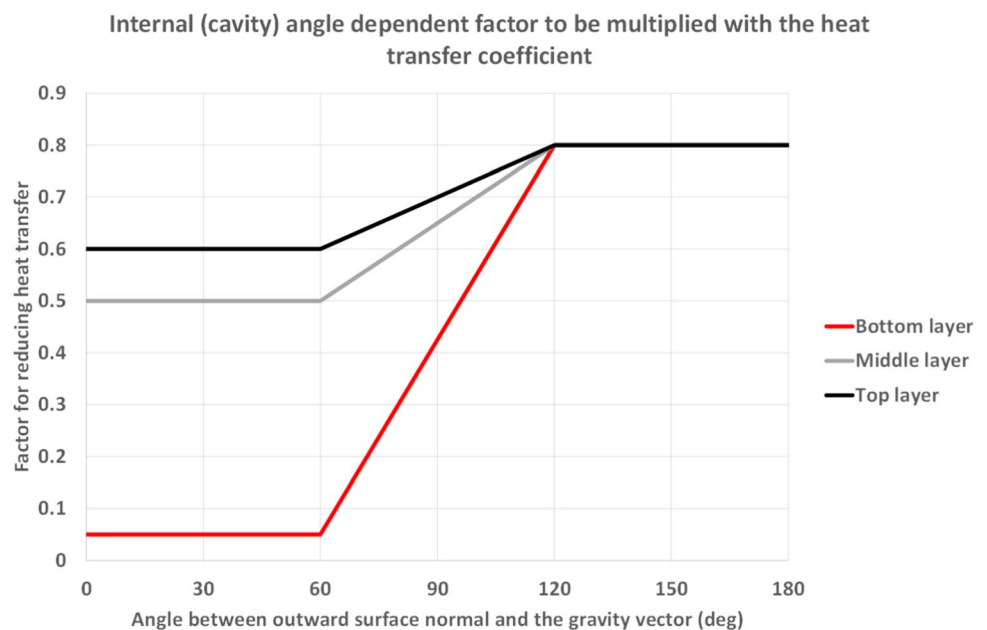
- AC deformation
- Creep-only heat treatment
- Heat treatment with coupled creep and quenching

3 Results and discussion

3.1 Results on AC deformation

This subsection presents the results and observations from the FE simulation conducted to predict the AC distortions during the casting process. This includes distortions

Fig. 11 Inside cavity (i.e. not external) angle dependent factor to be multiplied with the heat transfer coefficient from the vapour modified water cooling correlations



imposed by the cold mould cavity expansion, the heated mould and distortions due to casting, cooling and degating. The entire process is simulated with FEM using IFE's in-house software StaMiSim. The sandcore removal was not included in the simulation.

The casting program CastCad will simulate several cycles starting from a cold mould, and continue until a stationary situation appear, approximately after 10 cycles. The FE simulation with StaMiSim is then conducted including the stresses in the part and the mould equipment surrounding the part. Our simulation domain during the solidification is in contact with the surroundings at two boundaries. On the top of the tilt plate, which is the uppermost part of the picture in Fig. 12, two cylinders are pressing down with a pressure corresponding to about 100 tons. Even so the tilt plate and most of the other computational domains will move upwards when it is heated, at most until 4mm at the corners of the upper mould, this is due to the strong forces induced by thermal expansion. The other external boundary is the machine plate connections at the bottom of our computational domain in Fig. 12. This is at the top of the crucible which is considered fixed. A small negative vertical displacement is visible in the middle of the machine plate due to the pressure from above on the tilt-plate. Figure 12 shows the calculated vertical displacement on the equipment from the casting of an automotive subframe. The lower and the upper mould are fairly thin walled which minimizes the deformations. We can see that at the corners of the lower and upper mould there is a gap equal to about 1.5mm.

As the part is solidifying an airgap will in some places form due to shrinkage away from the mould surface (the thermal expansion coefficient for aluminium is higher than for the steel, so the aluminium will shrink faster during cooling and a gap might form). A non-linear mechanical boundary condition is used for that purpose which allows for an airgap to develop between the mould and the part and at the same time applies a penalty on penetration. Figure 13 shows the calculated airgap at 100 seconds from the start of the filling of the mould, the part is not fully solidified but some of the most thin-walled sections are already solid.

While during casting the part is resting on the contact points with the mould, after casting it is transported on a palette for sandcore knock-out and degating. New resting points are defined and therefore the part will move and rotate accordingly, but the displacements from the nominal geometry are not influenced by the rotation. Most of the AC deformations occur within the mould, but a smaller portion develops during degating due to the release of internal stresses. Figure 14 shows the displacements from the nominal geometry (at room temperature) after degating, while Fig. 15 shows a comparison between averaged data from test production together with the simulation results.

The main distortions during the casting is captured, taking into account the variations also in the measured data. The largest deficit in the modelling is that the thermal simulation conducted for this work does not include the data from the air-gap formation computed by the mechanical model.

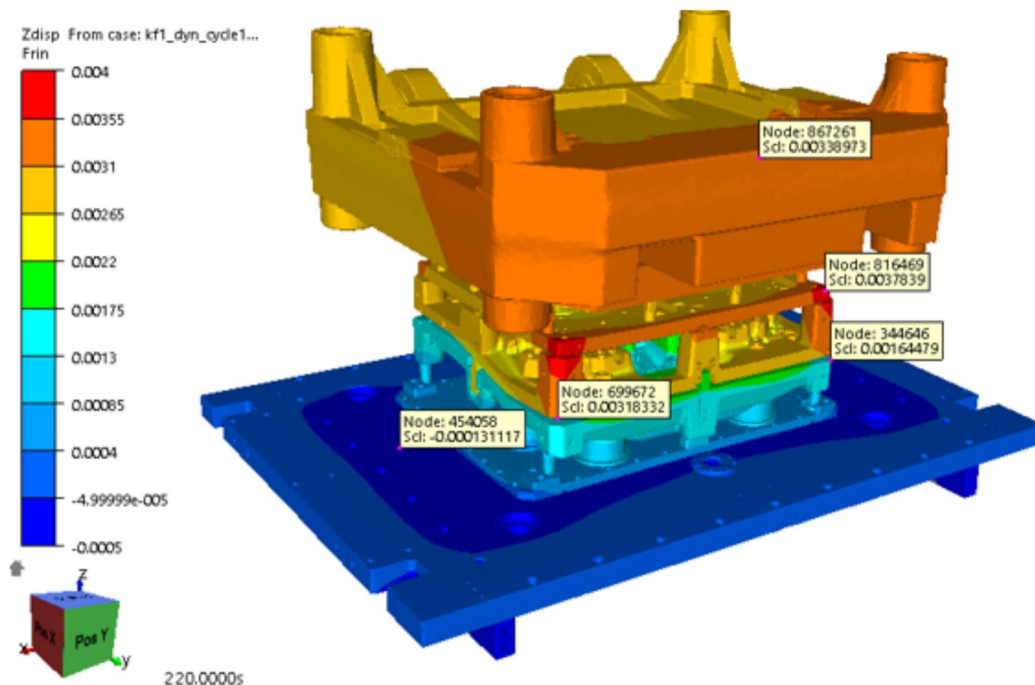


Fig. 12 Vertical displacements (unit meter) at selected points on the mould equipment. The deformed view is exaggerated 20 times (displacements multiplied with 20). The colour scale is from -0.5mm to 4mm

Fig. 13 Airgap size with colour scale from 0 (dark blue) to 0.6mm (red) in steps 0.06mm, at 100sec from start of filling the cavity

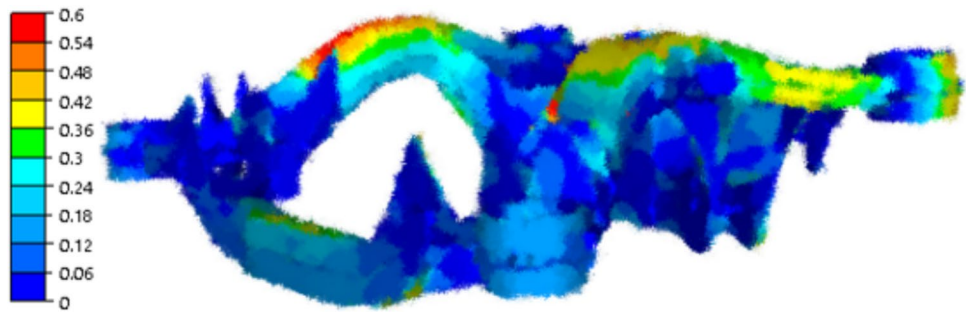


Fig. 14 As cast displacements from the nominal geometry on automotive subframe after slow cooling to room temperature. Colour scale is from -2mm to +2mm in steps 0.4mm from dark blue via green and yellow to orange and red

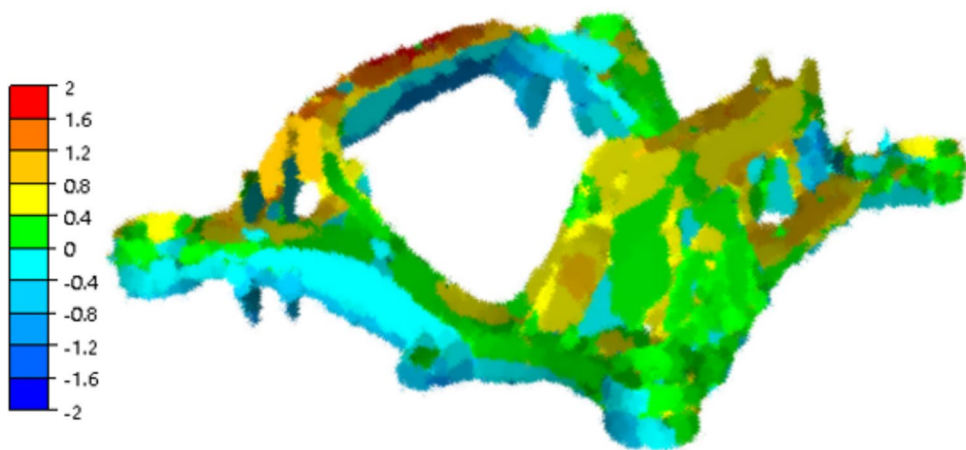
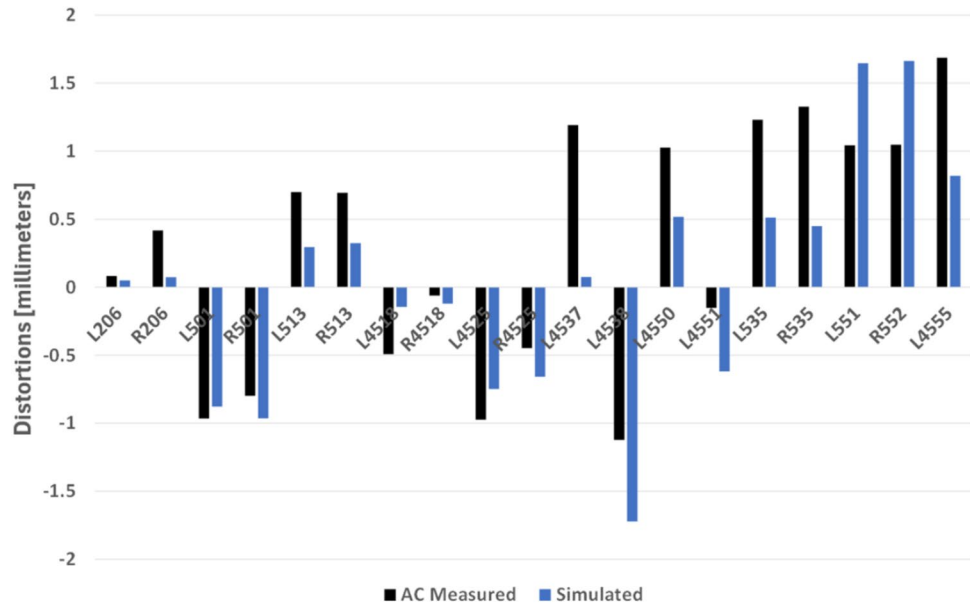


Fig. 15 Comparison between simulated and measured as cast displacements from the nominal geometry for selected points on a subframe. Unit is millimeter



3.2 Results on heat treatment with creep only

By measuring the AC deformations before heat treatment and subtracting these values from post-treatment measurements, we can determine the movement that occurs during heat treatment. This movement is primarily caused by quenching; however, depending on the part's weight and its positioning within the rack (i.e., the chosen support points),

significant creep due to gravity may also occur. By using only air cooling—which, based on our experience, does not cause significant movement provided the part is solid and free of cavities—we can assess the extent of creep. By comparing the simulation results with the experimental data we can validate the creep properties in the model.

Figure 16 shows the movement from creep only on the subframe in Fig. 5. The width of this subframe is about

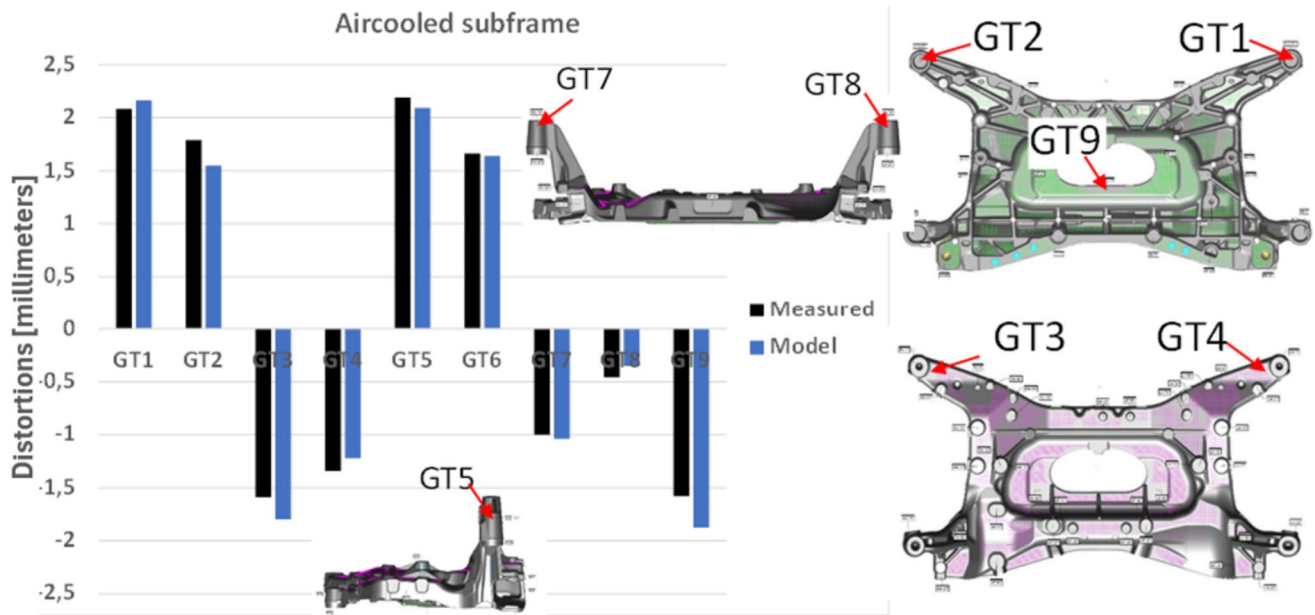


Fig. 16 Displacements from the nominal geometry in heat treatment due to creep, and compared with simulation

1m and the weight is about 18kg, but the bushing connections are heavy and located outside the support rods in the rack. Gravity forces will therefore act on the bushings and with the weak material at 540°C, they are moving slightly downwards.

3.3 Results on heat treatment with combined creep and quenching

The measured displacements from the nominal geometry on different automotive parts were used to calibrate the empirical water cooling correlations including water temperature increase, vapour amount and angle dependent factors. The weight of each rack including the cast parts was used to scale the increase in water temperature and vapour content in the tank—greater mass results in a higher temperature rise and increased vapour content. In addition the geometric shape is of course different for each part so the surface angles will differ as well. For the frame where we measured the as cast deformation as shown in Fig. 15; the measured movements in the first trials are shown in Fig. 17 and the simulated values in Fig. 18. In this example 3 layers (bottom, middle and top) were used in the rack, and as we can see there is a significant change between layer 1 (bottom layer) and the other two layers. The cooling for layer 1 is higher as it moves for a longer time through the water (i.e. more forced convection cooling) and less vapour will also be present around the lowest layer. Looking on the distortions we can see how the whole subframe wraps the other way going from the bottom layer with the highest deformation through the middle layer

(with small deformation) and into the top layer with larger deformations again.

However, for other cases the highest distortions made during heat treatment might appear for the higher levels, it depends on the geometry, the water temperature, the rack lowering speed etc. By using this knowledge a new rack could be constructed, which utilizes the creep movement, and thereby mitigates the movement during the quenching. The contact points with the rack were moved in order to increase creep for the first layer, which reduced the overall distortions after the heat treatment. In this way a high productivity rack with 5 layers were constructed with different contact points between rack and parts for each layer. The optimized results are shown in Fig. 19, using the same scale as in the first trials results in Fig. 17. There is a significant reduction in the overall heat treatment distortions, down to approximately 1mm. The simulations in Fig. 20 shows a similar reduction, but as the movements are reduced it is also more challenging for the model to capture all the details, as can be seen for the top layer which wraps the other way around in the model.

For the cross-member where we showed the internal cavity marker (Fig. 9) the measured movement is shown in Fig. 21 and the simulated values in Fig. 22. In this example 6 layers were used in the rack, and again, as observed above there is a significant difference between the layers.

The stacking of cross-members in the rack is shown in Fig. 23, note the positioning of each part in the rack. The positioning will decide which ways the parts will bend as explained below. The reason for the different results



Fig. 17 Distortions on subframes from different horizontal layers during heat treatment with quenching in a water tank. Experimental results

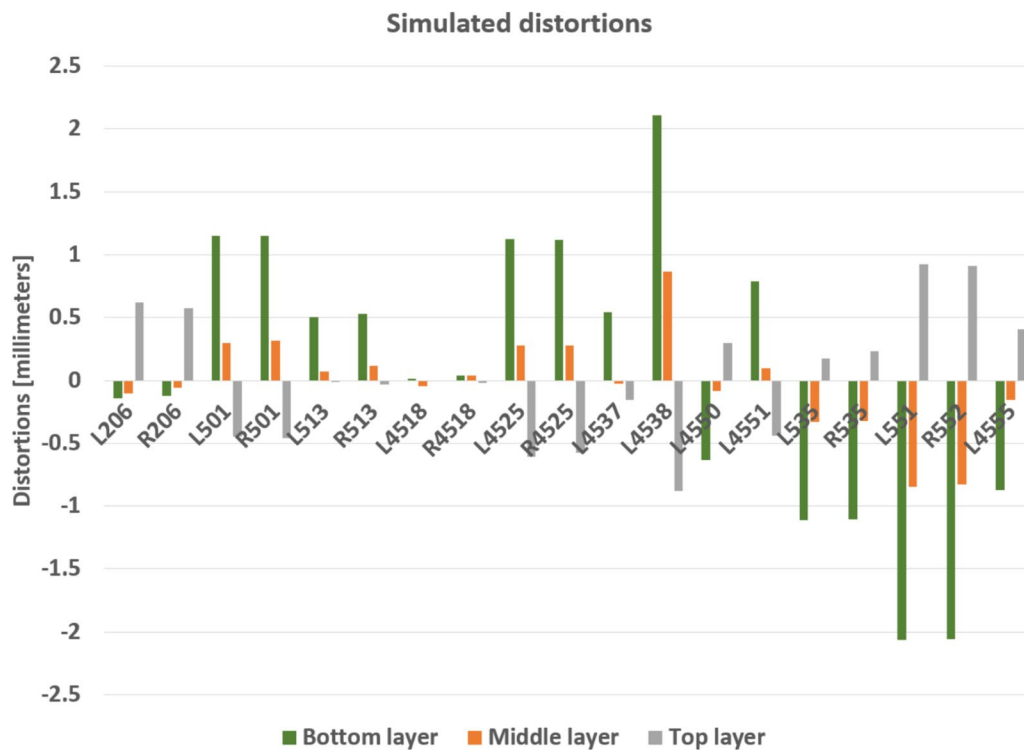


Fig. 18 Distortions on subframes from different horizontal layers during heat treatment with quenching in a water tank. Modelling results

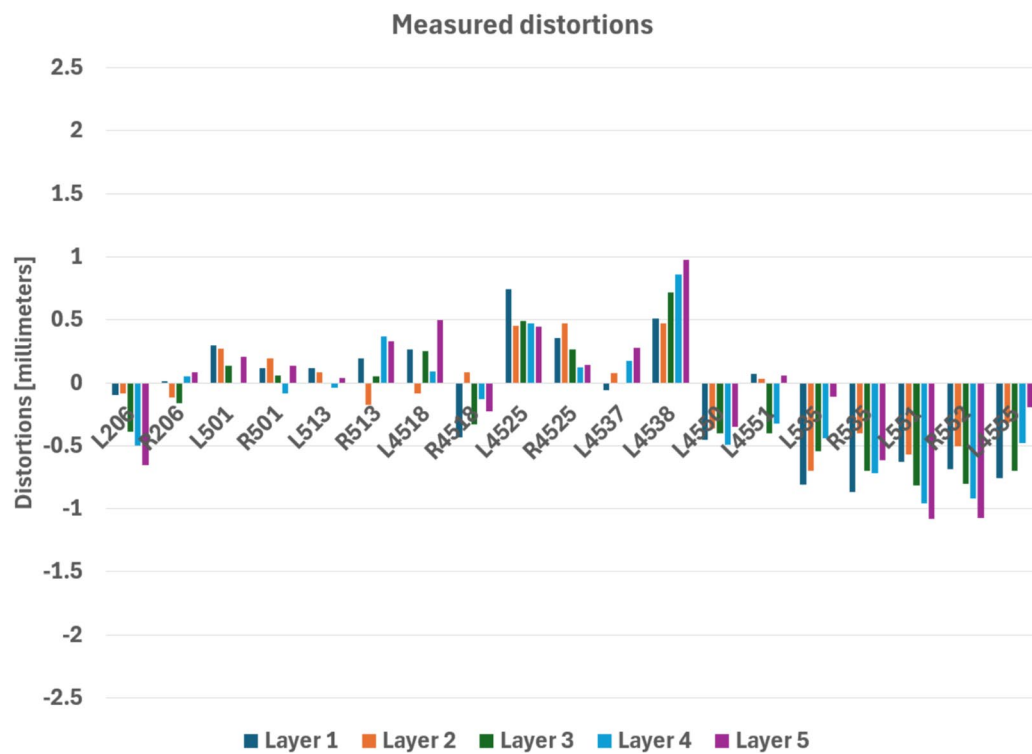


Fig. 19 Distortions on subframes from an optimized rack with five different horizontal layers during heat treatment with quenching in a water tank. Layer 1 is at the bottom and Layer 5 at the top. Experimental results

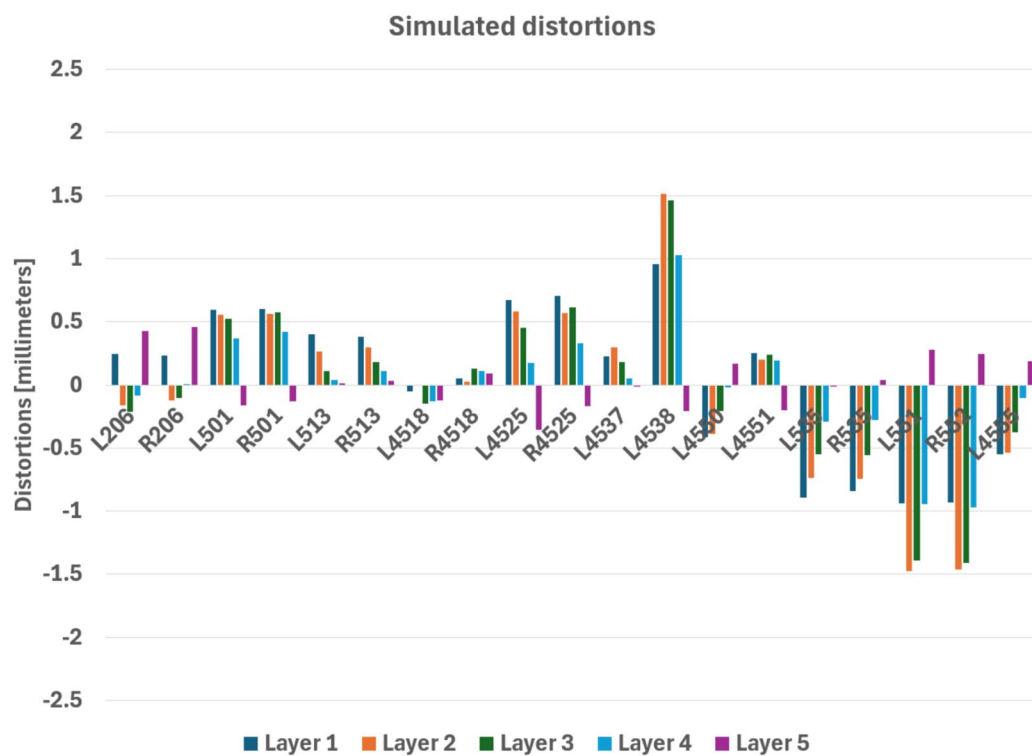


Fig. 20 Distortions on subframes from an optimized rack with five different horizontal layers during heat treatment with quenching in a water tank. Modelling results

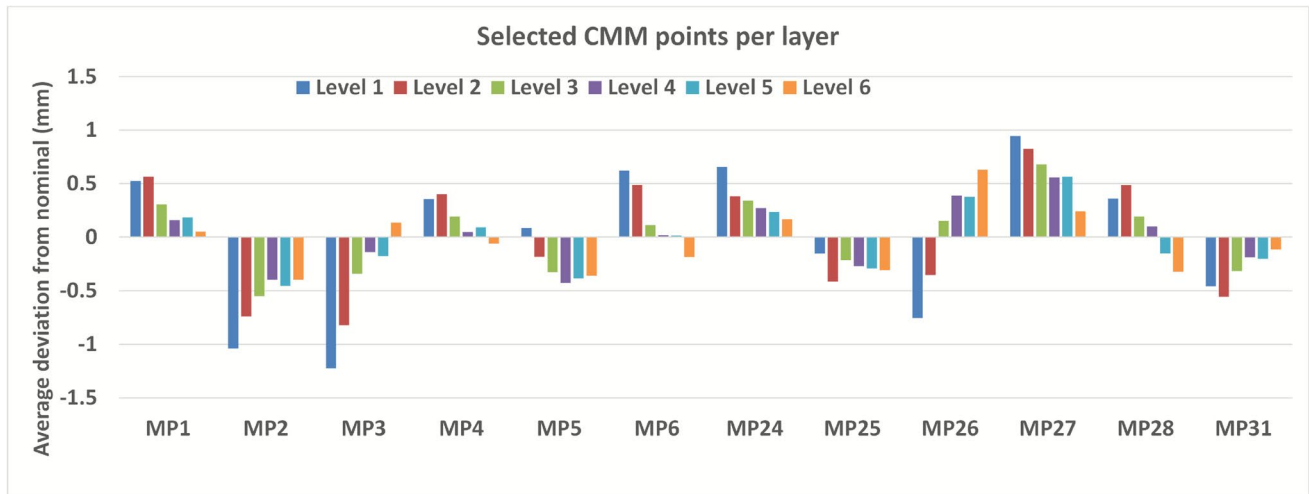


Fig. 21 Distortions on selected points on a cross-member during heat treatment with quenching in a water tank. There are six horizontal layers in the rack. All layers are shown for each point, starting with layer 1 which is blue (the bottom layer). Experimental results

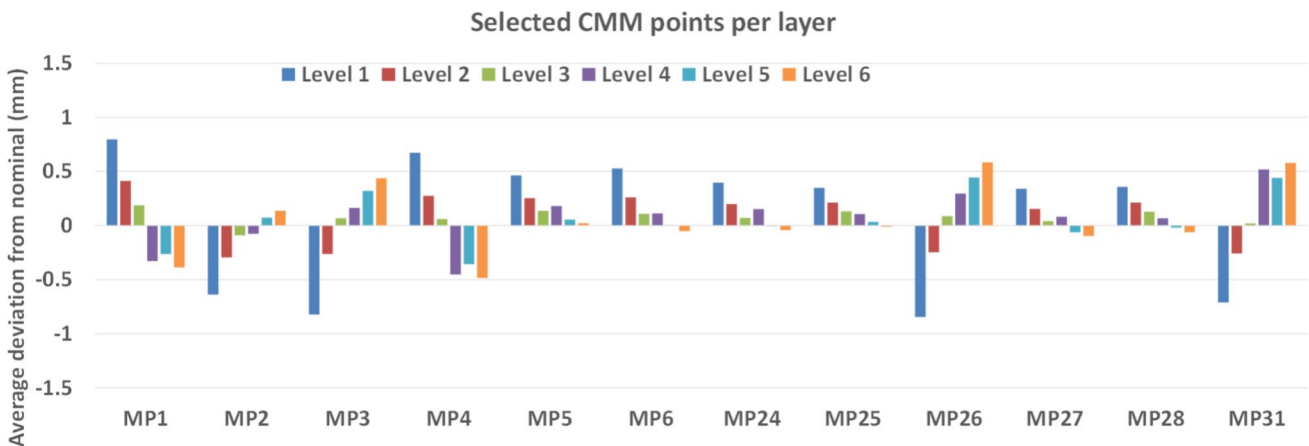


Fig. 22 Distortions on selected points on a cross-member during heat treatment with quenching in a water tank. Modelling results

between the layers are illustrated in Figs. 24 to 27. In Fig. 24 the upper part of the cross-member is still hot and therefore longer than the already cooled lower part of the beam – which is shorter. However, the upper part of the member “would like” to be even more longer but the extension is restricted by the cold and stronger lower part of the member. The upper (hot and weak) part is therefore compressed by the lower (cold and strong) part resulting in permanent plastic compression for the upper part of the beam. Therefore – when the part is fully cooled it will bend the other way as can be seen in Fig. 25.

In layer 1 the temperature difference along the cross-member is larger than for layer 6. The reason for this is that the cross-members in layer 1 will move for a longer time through the water, and thereby exposing the lower side of the member for a higher cooling rate. Inside the member there is a cavity and due to strong boiling in the very first few seconds the upper side of the cavity accumulates

vapour inside which cannot be released as fast as bubbles created on the outside of the member. This effect enlarges the temperature difference between the top and bottom of the member. However, for layer 6 this temperature difference is smaller as the cross-members there will travel only for a short time through the water and the empirical angle differences between downward facing surfaces and upward facing surfaces, as shown in Figs. 10 and 11, are smaller. As can be seen in Figs. 26 and 27 this results in less plastic compression and thereby smaller distortions for the fully cooled member.

As discussed in Section 2, the Bauschinger effect was neglected in the constitutive model based on the assumption that solution heat-treated A356 alloys exhibit only a weak Bauschinger response. However, bi-directional loading observed in the simulations suggests that this effect is indeed present, although its implications are not investigated further in the present study.

Fig. 23 Picture of test rack (including logging box at the upper left position) which illustrates the stacking of cross-members in six layers inside the rack

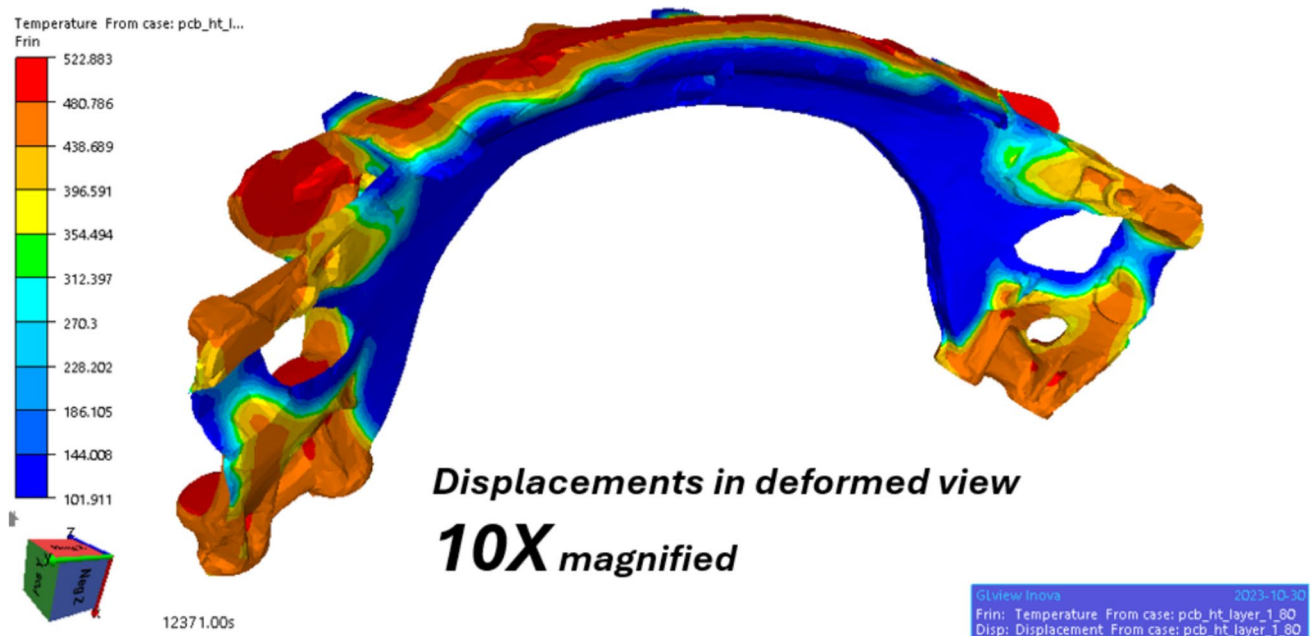


Fig. 24 Bending of cross-member in layer 1 after 2 seconds in the water tank. At this time the cross-members in layer 1 is still travelling downwards in the tank. The colour scale is from 102°C to 523°C

Modelling will give an understanding regarding the detailed mechanisms behind the quench distortions. This understanding can be used to mitigate the distortions. Ways of influencing the distortions are to vary the part positioning in the rack, the speed of the rack when it is lowered into the water, the water temperature and also imposed stirring of the water. Further on the electrical conductivity of the water (or water quality in general) will influence the water cooling, because especially film boiling is influenced by the water quality (T_{MFB} and q_{FB}) in Fig. 6.

As can be seen from the comparison with measured displacements from the nominal geometry we are capturing the main distortions with the modelling tool but not all the details. However, the most important usage will be to find out

how to influence the distortions, to move in the right direction in order to reduce the maximum deviations. Another application would be to let the quench distortions mitigate the AC deformation when possible. In order to increase the accuracy of the tool it has to be tested against as many as possible automotive parts in order to improve the empirical correlations. Specifically the internal cavity angle factor can be improved taking into account measures for how easy it is for the generated vapour bubbles to escape from different geometries. Using computational fluid dynamics on a single part can further refine the empirical factors in the modelling approach for the full rack. For the AC deformation, coupling the air-gap prediction with the thermal simulation will be an important improvement.

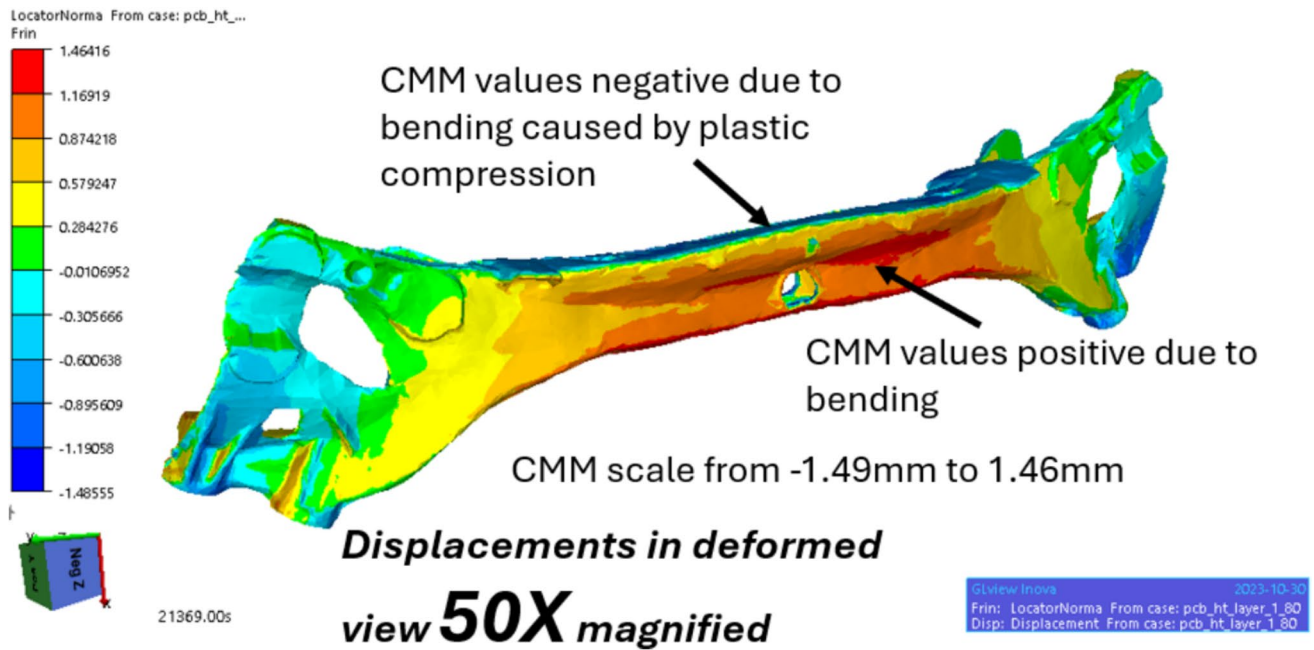


Fig. 25 Bending of cross-member in layer 1 after it is fully cooled

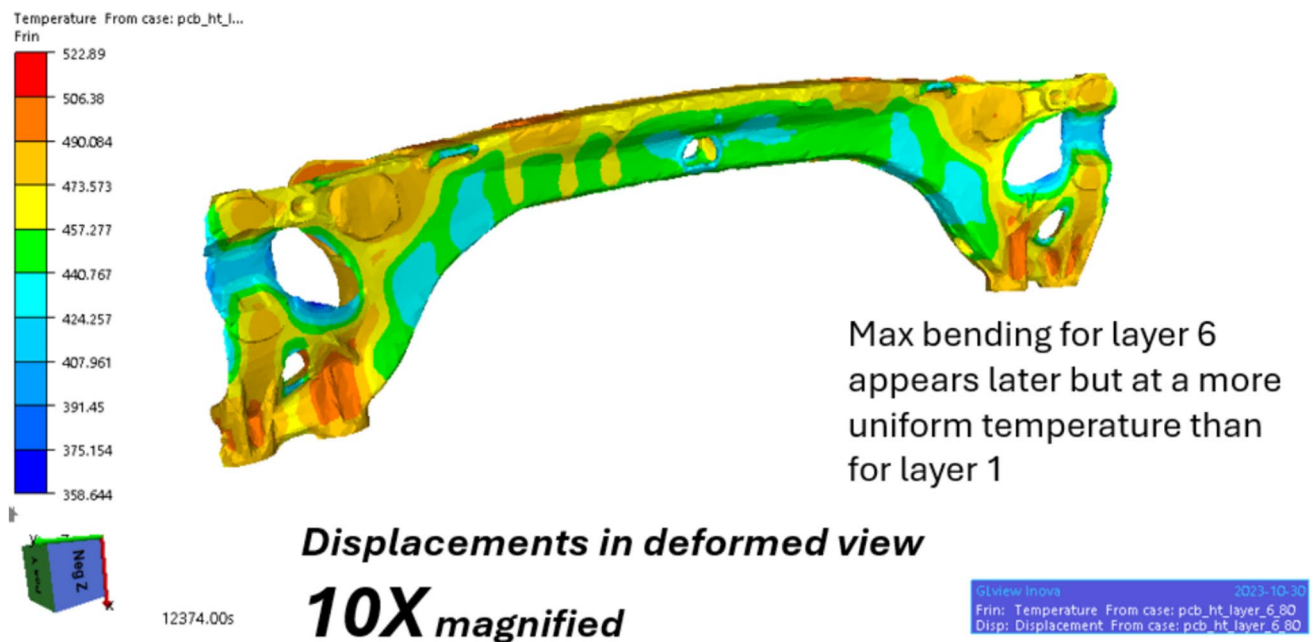


Fig. 26 Bending of cross-member in layer 6 after 2 seconds in the water tank. At this time the cross-members in layer 6 are done with travelling downwards in the tank. The colour scale is from 358°C to 523°C

4 Conclusion and next steps

This paper presents the modelling strategy adopted to predict distortions on large aluminium automotive subframes. The simulations capture the distortions due to solidification and mould deformation, heat treatment on racks and water quenching of the parts. The simulation results were compared with data from experiments measuring the distortions

from the physical process. The comparisons show a good correlation between the computed and test values; however there are some variations which have been discussed and possible improvements suggested to improve the comparison. The modelling methodology and assumptions adopted in this work will contribute to reducing the scrap rate, and also enabling the casting of new products covering a larger function in the car (replacing many smaller parts),

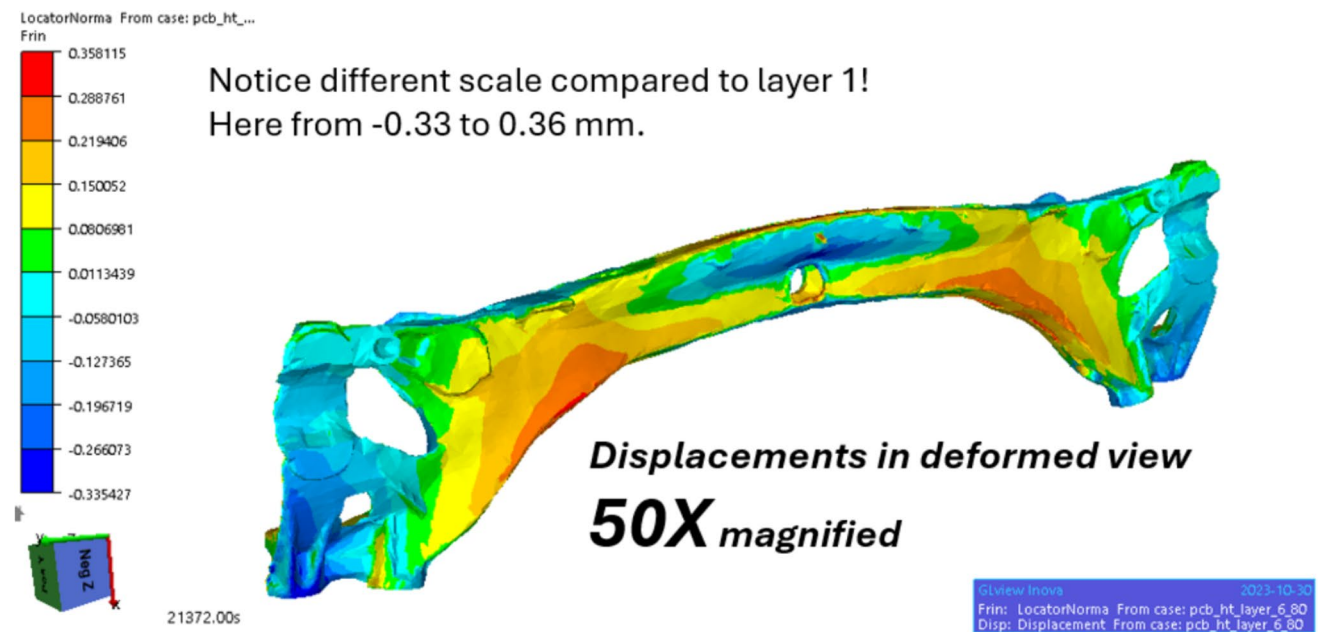


Fig. 27 Bending of cross-member in layer 6 after it is fully cooled

Table 3 Nomenclature

Symbol	Description
σ_s	Solid stress tensor
g_s	Solid volume fraction
g_l	Liquid volume fraction
p_l	Pressure in the liquid phase
ρ_s	Density of solid phase
ρ_l	Density of liquid phase
$\mathbf{g}$	Gravity vector
$\boldsymbol{\varepsilon}$	Total strain tensor
$\mathbf{u}$	Displacement vector
$\boldsymbol{\varepsilon}^{vp}$	Viscoplastic strain component
$\boldsymbol{\varepsilon}^e$	Elastic strain component
$\boldsymbol{\varepsilon}^T$	Thermal strain component
β_T	Volumetric thermal expansion coefficient
T	Temperature
T_{coh}	Temperature at coherent mushy zone boundary
$\mathbf{I}$	Identity matrix
$\mathbf{D}(T)$	Elasticity matrix (temperature-dependent)
$\boldsymbol{\sigma}$	Stress tensor (general)
ρ_s	Density of solid material
$\bar{\sigma}$	Effective stress
$\dot{\bar{\varepsilon}}_p$	Effective viscoplastic strain rate
$F(T)$	Temperature-dependent material function
ϕ	Work hardening parameter
ϕ_0	Work hardening constant
$n(T)$	Temperature-dependent hardening exponent
$m(T)$	Temperature-dependent strain rate exponent
T_0	Temperature below which there is work hardening

supporting sustainable design goals within automotive production.

The models and discussion in this work also offer ideas for next steps in the improvement of correlations between the experimental and simulation results. The water cooling description defined in this work depends on the empirical calculation of the temperature in the tank and some other assumptions (based on engineering judgement). With data from sensors these calculations can be improved with a mathematical model giving a more robust and reliable model. Similarly, the vapour bubble correlation factor was developed purely by an empirical approach and could be further explained by mathematical modelling. Bi-directional loading was observed in the heat-treated component in the simulations, indicating the presence of a Bauschinger effect; consequently, incorporating a back-stress term into the constitutive equations could improve the predictive capability of the model. Additionally, generating more data from physical tests including cyclic loading will enhance the comparison and give a better outlook on the limitations of modelling the process.

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Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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